UNIVERSITY EXAMINATIONS: 2014/2015

ORDINARY EXAMINATION FOR THE BACHELOR OF BUSINESS INFORMATION TECHNOLOGY

BBIT 106 DATA STRUCTURES AND ALGORITHIMS - DISTANCE LEARNING

DATE: DECEMBER, 2014 **TIME: 2 HOURS**

INSTRUCTIONS: Answer Question ONE and any other TWO

QUESTION ONE [30 MARKS]

a) Describe the steps for designing an algorithm. [5 Marks]

b) Differentiate between a walk through and a desk check as used in testing the algorithm. [4 Marks]

c) Describe any three operations performed in a binary tree data structure. [6 Marks]

d) Design an algorithm that inserts a node at the beginning of a linked list. [5 Marks]

e) Design an algorithm that uses two nested **FOR** loops to print the pattern below:

```
*
*   *
*   *   *
*   *   *   *
*   *   *   *   *
*   *   *   *   *   *
*   *   *   *   *   *   *
*   *   *   *   *   *   *   *
```

[6 Marks]

[a]. Describe linear probing as one of the techniques for solving collisions in a hash table. [4 Marks]

QUESTION TWO [20 MARKS]

(a) Design an algorithm using a pseudo code that uses the **SWITCH** keyword to prompt the user to enter a grade then display the remarks using the criteria below.

Grade	**Remark**
1	Agree
2	Strongly agree
3	Disagree
4	Strongly disagree
5	Neither agree nor disagree

[8 Marks]

(b) Calculate the big O for the in each of the following functions:

[i]. $f(n) = \frac{3}{n^4} + 6n^3$

[ii]. $f(nm) = 2(n - m) + 6nm^3$

[iii]. $f(n) = (3n+2)^3$

[6 marks]

(c) Briefly describe why algorithm analysis is necessary. [6 Marks]

QUESTION THREE [20 MARKS]

a) Briefly explain how array elements can be accessed. [6 Marks]

b) Explain the difference between static and dynamic arrays. [6 Marks]

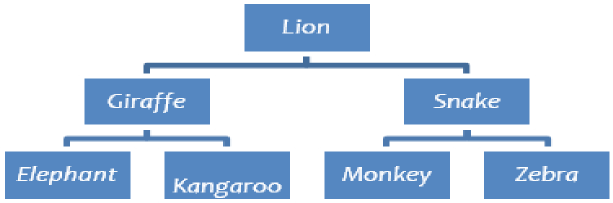
c) Design an algorithm using a pseudo code to calculate and display average score for forty scores keyed in by the user. [8 Marks]

QUESTION FOUR [20 MARKS]

a) Define a queue ADT and hence explain any three operations of a queue. [6 marks]

b) Define a stack ADT and hence outline any applications of a stack ADT. [8 Marks]

c) A binary tree is a tree in which each parent node can have a maximum of two child nodes. Each node has two pointers- left pointer and right pointer. Given a binary tree below, you are required to display its elements using in-order and preorder traversals.



[6 Marks]

QUESTION FIVE [20 MARKS]

a) Design an algorithm using pseudo code to sort elements by insert sort technique. [8 Marks]

b) Design an algorithm using pseudo code to search elements in an array of size n by linear search. [8 Marks]

c) Briefly explain the two techniques of traversing a graph ADT. [4 Marks]